EFU DEFENSE LICENCE v1.0

Experimental EFU-based Defense & Kinetic Energy Framework

0. Purpose and Scope

Experimental EFU-based licensing and measurement framework for defense (military, weapon systems, logistics, kinetic energy).

NOT a military secrecy regulation, defense procurement permit, or operational command replacement, but an open research protocol measuring defense's physical entropy cost in human-scale EFU metrics (fuel, ammunition, maintenance, explosives). Covers land, sea, air forces, logistics.

I. Core Concepts

1. **EFU-Defense (EFU-DEFENSE)**
   EFU framework application for kinetic operations:
   - **EFU-F_DEF**: fuel (tanks, aircraft, ships)
   - **EFU-A_DEF**: ammunition, explosives, missile propellants
   - **EFU-M_DEF**: weaponry, armor, vehicle maintenance
   - **EFU-E_DEF**: base energy, communications, radar centers
   - **EFU-K_DEF**: kinetic entropy (exercises, operations)
2. **Defense Sovereignty Gap (SS_DEFENSE)**
   SS_DEFENSE = TEFU_DEFENSE_total – Regional Defense Carrying Capacity (RDCC_DEFENSE)
   Measures if defense stays within national territory's regenerative capacity.
3. **MROI_DEFENSE**
   MROI_DEFENSE = THI_DEFENSE / TEFU_DEFENSE
   THI_DEFENSE = territory protection + sovereignty preservation + peacetime efficiency.

II. Licence Status

Documents EFU-audited defense ministries, armament programs.
Why critical: Defense = **highest entropy/sovereignty ratio** globally.
Enables Swiss militia vs. US base network benchmarking.

III. Metrics and Thresholds

1. EFU Components (per national territory km²)

EFU-F_DEF (Fuel)
EFU-F_DEF = fuel_MWh / 12.88 / 1000 km² territory
Example: USA = 50B MWh/9M km² → 0.55 EFU/km²

EFU-A_DEF (Ammunition/explosives)
EFU-A_DEF = TNT_equivalent_tonnes / 1t/person/year / 1000 km²

EFU-K_DEF (Kinetic entropy)
EFU-K_DEF = training_hours × platforms / 100 hours/km²/year

TEFU_DEFENSE = 2.0×EFU-F + 2.5×EFU-A + 2.2×EFU-M + 1.5×EFU-E + 3.0×EFU-K

2. Regional Defense Carrying Capacity (RDCC_DEFENSE)

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RDCC_DEFENSE = Sustainable_km² / Territory / 0.01 km²/km²/year
Example: SWITZERLAND = 41k km² protected / 41k = 1.0 EFU_DEFENSE/km²/year
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IV. Licence Categories (EFU-DEFENSE Labels)

Level	SS_DEFENSE	MROI_DEFENSE	Label	Interpretation
1	≤ 0	≥1e-1	**EFU-DEF SOVEREIGN**	Militia/wire fence
2	0 < ≤50k EFU	≥5e-2	**EFU-DEF BALANCED**	Regional defense
3	50k < ≤250k EFU	≥1e-2	**EFU-DEF DEFICIT**	NATO Article 5
4	>250k EFU	<1e-2	**EFU-DEF CRITICAL**	Global military entropy

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V. Regional RDCC_DEFENSE Calibrations

Country	Territory (k km²)	Protected (k km²/year)	RDCC_DEFENSE
SWITZERLAND	41	41	1.00
FINLAND	338	300	0.89
ISRAEL	22	22	1.00
USA	9,800	3,000	0.31
RUSSIA	17,000	1,000	0.006

VI. Example: US Defense Entropy

Input: 9.8M km², 50B MWh fuel, 1M tonnes ammunition, 800 bases, 10M training hours

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EFU-F_DEF = 50B/12.88/9.8M = 0.55 EFU/km²
EFU-K_DEF = 10M hours/100/9.8M = 0.01 EFU/km²

TEFU_DEFENSE_total ≈ 75,000,000 EFU/year
Allocated = 9.8M × 0.31 = 3M EFU/year
SS_DEFENSE = +72M EFU/year (Level 4: CRITICAL)

Result: EFU-DEF CRITICAL
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Swiss benchmark: -0.2M EFU/year (Level 1: SOVEREIGN)

VII. Implementation Notes

1. **NATO 2% GDP**: TEFU_DEFENSE physical correlation
2. **Arms export**: EFU-A_DEF proliferation metric
3. **Peace missions**: MROI_DEFENSE peace/entropy ratio
4. **Data sources**: Tier 1 = fuel bills (public); Tier 2 = SIPRI military spending